MASTER FUNCTIONAL SKILLS

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AQA · LEVEL 1 · MATHEMATICS

AQA Functional Skills Mathematics Level 1

January 2019 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-FSM-L1-Jan19

100% Built from Empty Question Paper + Official Mark Scheme

Verified by Tutor Abdul · MasterFunctionalSkills.co.uk

HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Level 1
Data Book (Examination)

January 2019

Insert

Instructions
- This copy of the Data Book is for use in the examination. It should not be given to students in advance.

Advice
- This book will not be collected in for marking. Ensure that all working that you wish to have marked is written in the space provided in the Question Paper/Answer Book.

4367

QAN 500/8703/4

IB/M/Jan19/E4

Data Sheet for Part-time work

Tram timetables

Queens Road to Piccadilly Gardens					
Trams are at these minutes past each hour	01	13	25	37	49
The journey time from Queens Road to Piccadilly Gardens is 11 minutes					

Piccadilly Gardens to Queens Road					
Trams are at these minutes past each hour	04	16	28	40	52
The journey time from Piccadilly Gardens to Queens Road is 10 minutes					

Bus timetables

Piccadilly Gardens to Trafford Centre								
Piccadilly Gardens	0715	0745	0815	0845	0912	0932	0952	1012
Deansgate Station	0722	0753	0823	0853	0920	0940	1000	1020
Old Trafford	0732	0803	0833	0903	0930	0950	1010	1030
Trafford Centre	0747	0818	0848	0918	0945	1005	1025	1045

Trafford Centre to Piccadilly Gardens								
Trafford Centre	1733	1753	1813	1833	1853	1913	1933	1953
Old Trafford	1748	1808	1828	1848	1908	1928	1948	2008
Deansgate Station	1759	1819	1839	1859	1919	1937	1957	2017
Piccadilly Gardens	1806	1826	1846	1906	1926	1943	2003	2023

The National Minimum Wage

Workers should be paid at least the National Minimum Wage.

This table shows the hourly rates for different ages.

Age (years)	16 to 17	18 to 20	21 to 24	25 or over
Hourly rate	£4.20	£5.90	£7.38	£7.83

Data Sheet for On the farm

Dairy farm

A dairy farm is a farm where cows are kept for their milk.



The amount of milk a cow produces is called the **milk yield**.

This amount is measured in litres.

In the UK, the mean milk yield per cow per year is 7944 litres.

Farmers usually sell the milk for between 30p and 34p per litre.

Keeping pigs

Pigs are usually kept in buildings called **sties**.
Each **sty** should have a covered area to keep the pigs dry.
Each sty should have a gate at one end to allow access to open space.
Pigs like to have space to run around and digging areas to look for food.



A mother pig is called a sow.

Baby pigs are called piglets.

The table shows the minimum area of sty needed for sows and their piglets.

	Total number of pigs in the sty	
	5 or fewer	**6 or more**
Sow	2.48 square metres each	2.25 square metres each
Piglet	1.80 square metres each	1.64 square metres each

END OF DATA

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Answer **all** questions in the spaces provided.

1 **Part-time work**

There is a **data sheet** for Part-time work.

Emmie is a student.

She lives near Queens Road.

Q1 Worked Solution:
See official AQA mark scheme method.
[Method & Accuracy]



Emmie

Here are the details of Emmie's journey to work.

5-minute walk from home to the Queens Road tram stop
Get the tram from Queens Road to Piccadilly Gardens
Get the bus from Piccadilly Gardens to the Trafford Centre
10-minute walk from the bus stop at the Trafford Centre to the shop

Do not write outside the box

Emmie starts work at 10.00 am

She leaves home at 8.30 am to catch the tram at Queens Road.

The tram is on time.

1 (a) What time does she get to Piccadilly Gardens?

[2 marks]

1 (b) Emmie says,

"I should arrive at the shop before 10.00 am"

Is she correct?

You **must** show your working.

[5 marks]

Do not write outside the box

1 (c) One day, Emmie is paid £41.30

She pays these fares.

Queens Road to Piccadilly Gardens	£4.00 return
Piccadilly Gardens to the Trafford Centre	£3.70 return

Emmie says,

"After paying for fares I have **more than** £35 of my pay left over."

Is she correct?

You **must** show your working.

[4 marks]

†1 (d) Gail works with Emmie.

She is 21 years old.

She is paid the National Minimum Wage.

She works for 8 hours.

How much is she paid?

[2 marks]

Check your answer.

Show how you have done your check.

[1 mark]

14

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2 **On the farm**

There is a **data sheet** for On the farm.



I am a farmer.
I keep cows and pigs.

Ken

2 (a) One day, Ken records the milk yields, in litres, for 10 of his cows.

20.82	22.03	22.24	21.42	21.36
21.91	22.14	23.12	21.81	21.55

The UK mean milk yield per cow is 21.76 litres per day.

Ken says,

"The mean milk yield for these 10 cows today is **more than** 21.76 litres."

Is he correct?
You **must** show your working.

[4 marks]

Q2 Worked Solution:

See official AQA mark scheme method.

[Method & Accuracy]

Question 2 continues on the next page

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2 (b) In total, Ken has 150 cows.

The mean milk yield per cow is 7944 litres per year.

Ken sells all the milk for 33p per litre.

This year, his total costs for feeding and looking after the cows are £333 600

He says,

"This year, my profit on the milk from these cows is **more than** £60 000"

Is he correct?
You **must** show your working.

[5 marks]

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†2 (c) Ken wants to build a sty for one sow and one piglet.
Work out the minimum area of sty he needs to build.

[2 marks]

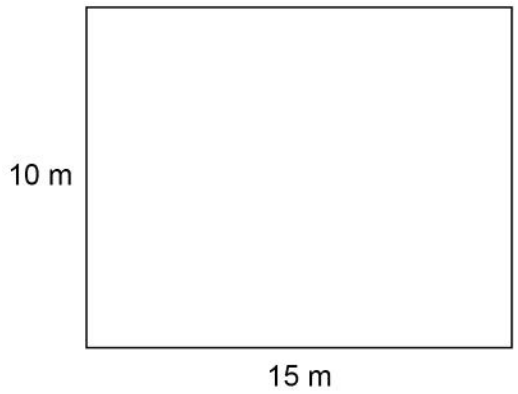
Check your answer.
Show how you have done your check.

[1 mark]

Question 2 continues on the next page

Do not write outside the box

2 (d) Here is a sketch of the fence around some land that Ken will use for new sties.



Not drawn accurately

Ken wants five square sties around the edges of the land, each 4 m by 4 m

The rest of the land will be open space.

Each sty must have at least one side facing the open space.

In the open space he wants

- a rectangular digging area of at least 30 m^2
- a rectangular feeding trough, 2 m by 3 m

Show a possible design for the land.

[3 marks]

Practise on this grid.

Scale 1 cm represents 1 m

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3 **Log burner**

Rohan wants to put a log burner in his living room.



Here is some information about three log burners.

The log burners are called the Brunel, the Chester and the Dover.

		Dimensions of log burner (mm)		
	Heat output (kW)	**Height**	**Width**	**Depth**
Brunel	4	524	385	290
Chester	6	592	553	397
Dover	11	678	759	403

3 (a) Rohan says,

"The width of the Dover is **more than** double the width of the Brunel."

Is he correct?

You **must** show your working.

[3 marks]

Q3 Worked Solution:

Marks are awarded for demonstrating the following interrelated process skills.

Representing Selecting the mathematics and information to model a situation.

R.1

Candidates recognise that a situation has aspects that can be represented using mathematics.

R.2

Candidates make an initial model of a situation using suitable forms of representation.

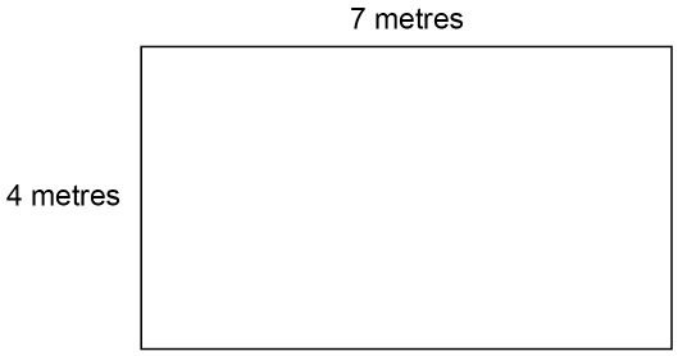
R.3

Candidates decide on the methods, operations and tools, including ICT, to

[Method & Accuracy]

Do not write outside the box

3 (b) The height of Rohan's living room is 3 metres.
Here is a floor plan of the room.



Not drawn accurately

Rohan uses these steps to work out the heat output needed for the room.

Step 1 Work out length of room × width of room

Step 2 Work out answer to **Step 1** × height of room

Step 3 Work out answer to **Step 2** ÷ 14

What is the smallest log burner with enough heat output for this room?
You **must** show your working.

[5 marks]

Question 3 continues on the next page

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3 (c) Dania has a log burner.

The log burner can burn smokeless fuel or wood.

She finds these costs.

- 16 kg bag of smokeless fuel costs £7.50
- 12 kg bag of wood costs £6.90

Dania's log burner uses 2 kg of smokeless fuel or wood per hour.

She uses the log burner for 24 hours per **week**.

Dania says,

"Using smokeless fuel is cheaper than using wood by **more than** £5 per week."

Is she correct?

You **must** show your working.

[7 marks]

Do not write outside the box

4 **Apricot jam**

Here is a recipe for apricot jam.

Ingredients for one batch of jam	
Apricots	2 kg
Sugar	600 g
Water	250 ml

Method

Put the ingredients into a pan

Heat the mixture to 105 °C

Cook for 40 minutes

4 (a) Convert 250 ml to litres.

Circle your answer.

[1 mark]

0.025 litres 0.25 litres 2.5 litres 25 litres

Q4 Worked Solution:

4 (b) Peter is going to use this recipe to make apricot jam.

He has

8 kg of apricots

2500 g of sugar.

He says,

"I have enough sugar to make the jam using **all** 8 kg of apricots."

Is he correct?

You **must** show your working.

[4 marks]

In particular, individual marks are mapped onto the following skills standards.
Representing
Making sense of the situations and representing them.
A learner can:
Understand routine and non-routine problems in familiar and unfamiliar contexts and situations.
Identify the situation or problems and identify the mathematical methods needed to solve them.
Choose from a range of mathematics to find solutions.
Analysing
[Ra] [Rb] [Rc] [Aa] [Ab]

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Peter starts to heat the mixture.

4 (c) This thermometer shows the temperature of the mixture after a few minutes.



How many degrees hotter does the mixture need to be to reach 105 °C?

[2 marks]

4 (d) The temperature reaches 105 °C at 3.35 pm

At what time should Peter finish cooking the jam?

[2 marks]

4 (e) Peter makes 20 pounds of jam.

1 pound = 16 ounces

He puts the jam into jars.

Each jar holds 10 ounces of jam.

How many jars can Peter fill with jam?

[3 marks]

Do not write outside the box

4 (f) Sandra also makes apricot jam.

She makes 25 jars of jam.

Each jar of jam costs £3.04 to make.

She sells each jar of jam for £3.99

She says,

"I have made a profit of **more than** £24"

Is she correct?

You **must** show your working.

[4 marks]

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END OF QUESTIONS

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Question number	**Additional page, if required.** **Write the question numbers in the left-hand margin.**

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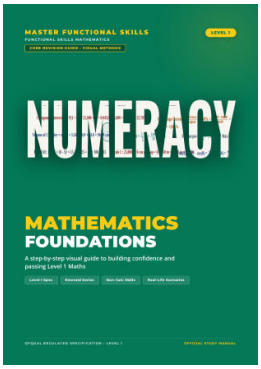
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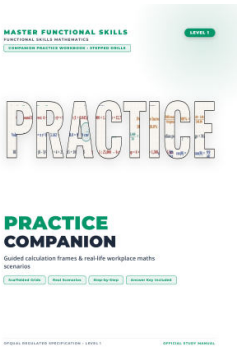


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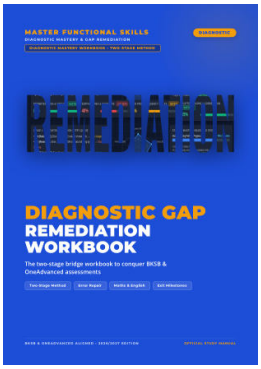


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